

① Sentence :-

* chatbot can answer user questions.

Tokenization :-

Splitting a sentence into individual words or tokens.

[chatbot, can, answer, user, questions]

embedding :-

[chatbot, can, answer, user, question]

* chatbot — [0.12, 0.45, 0.78]

* can — [0.34, 0.21, 0.56, 0.29]

* answer — [0.67, 0.89, 0.23]

* user — [0.43, 0.76, 0.91]

* question — [0.55, 0.28, 0.72]

② Sentence - 2 :-

* I am Priya

Tokens :- [I, am, priya]

embedding :-

* I — [0.36, 0.80, 0.90, 0.38]

* am — [0.34, 0.21, 0.56]

* priya — [0.67, 0.89, 0.23]

③ Sentence-3:- RNN Reads words

Tokens:- [RNN, Reads, words]

embedding:-

* RNN - [0.34, 0.21, 0.56, ...]

* Reads - [0.67, 0.89, 0.23]

* words - [0.45, 0.55, 0.78]

④ Sentence-4:- cohort team-4

Tokens:- [cohort, team-, 4]

embedding:-

* cohort - [0.26, 0.27]

* Team - [0.34, 0.39]

* 4 - [0.29, 0.46, 0.48]

⑤ Sentence-5:- we are students

Tokens:- [we, are, students]

embedding:-

* we - [0.36, 0.49, 0.58]

* are - [0.78, 0.10, 0.12]

* students - [0.57, 0.68]

⑥ Sentence-6:- I like AI

Tokens:- [I, like, AI]

embedding:-

* I - [0.12, 0.49]

* like - $[0.39, 0.46, 0.49]$

* AI - $[0.38, 0.56]$

⑦ Sentence-7: This is my project

Tokens: $[This, is, my, project]$

embedding:

* This - $[0.49, 0.57, 0.60]$

* is - $[0.80, 0.12, 0.25]$

* my - $[0.21, 0.48]$

* project - $[0.48]$

⑧ Sentence-8: She reads books

Tokens: $[she, reads, books]$

embedding:

* she - $[0.46, 0.59]$

* reads - $[0.80, 0.38, 0.48, 0.25]$

* books - $[0.90, 0.49, 0.12]$

⑨ Sentence-9: AI is useful

Tokens: $[AI, is, useful]$

embedding:

* AI - $[0.90, 0.82, 0.23]$

* is - $[0.60, 0.76]$

* useful - $[0.9, 0.46, 0.39]$

⑩ Sentence - 10 :- I am from G. Mamidada

Token :- [I, am, from, G. mamidada]

Embedding :-

* I - [0.30, 0.89]

* am - [0.12, 0.49, 0.83]

* from - [0.26, 0.76]

* G. mamidada - [0.39, 0.48, 0.57]

RNN - Recurrent neural network :-

① It Process one by one word read

* I am priya

⇒ Read the word "I"

⇒ Then reads the word "am"

⇒ Then reads the word "priya"

* Then I + am + priya = Introduction of person

Transformer Process :-

* The transformer is whole sentence read once

* I am priya

⇒ It understands that "I" refer to the Person speaking

⇒ It recognizes that "Priya" is the name of the person.

* It quickly understand the complete

meaning of the sentence

② Sentence - 2

chatbots can answer user questions.

RNN:- step by step read

* chatbots

↓

* can

↓

* answer

↓

* user

↓

* questions

Op:- Transformer

[chatbots] [can] [answer] [user] [questions]

Attention mechanism:-

* answer $\hookrightarrow$ question

* user $\leftrightarrow$ questions

* relation direct ga identify chatbots?